

SAHTY

Health Monitoring Platform

API Documentation

Complete REST API Reference Guide

VERSION

1.0.0

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Base URL: <http://localhost:8000>

Django REST Framework | Python 3.13 | PostgreSQL

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01

Authentication

User registration, login, email/phone verification, password reset, and account management. All authentication endpoints are public (AllowAnonymous) except delete_account which requires IsAuthenticated.

BASE URL	/users/auth/
----------	--------------

POST

/users/auth/signup/

Register a new user account. Sends a verification code to the provided email address.

Request Body (JSON)

Name	Type	Required	Description
first_name	string	Yes	First name (letters only, max 25)
last_name	string	Yes	Last name (letters only, max 25)
email	email	Yes	Unique email address
password	string	Yes	Account password
phone	string	Yes	Unique phone number (+XXXXXXXXXX)
birth_date	date	Yes	Date of birth (YYYY-MM-DD)
gender	string	No	M or F (default: F)
role	string	No	P (Patient) or D (Doctor), default: P

Success Response

200	{"success": true, "message": "Code sent successfully", "user_id": 1}
-----	--

Error Responses

400	Phone number already used / Email already used / Validation errors
500	Email not sent (SMTP failure)

POST **/users/auth/signin/**

Authenticate a user and return JWT access & refresh tokens.

Request Body (JSON)

Name	Type	Required	Description
email	email	Yes	Registered email address
password	string	Yes	Account password

Success Response

200 {"success": true, "access": "<jwt_token>", "refresh": "<refresh_token>"}

Error Responses

400 Invalid email or password
403 Email not verified / User did not complete signup

POST **/users/auth/resend_code/**

Resend the email verification OTP code to an existing user.

Request Body (JSON)

Name	Type	Required	Description
email	email	Yes	Registered email address

Success Response

200 {"success": true, "message": "Code sent successfully", "user_id": 1}

Error Responses

404 User not found
500 Email not sent

POST

/users/auth/verify_code/

Verify the email OTP code to activate the user account.

Request Body (JSON)

Name	Type	Required	Description
email	email	Yes	Registered email address
code	string	Yes	6-digit verification code

Success Response

200

{"success": true, "message": "User verified successfully"}

Error Responses

400

Code expired (new code sent automatically) / Incorrect verification code

404

User not found

POST

/users/auth/verify_reset_code/

Verify the OTP code for password reset flow. Sets can_reset_password flag.

Request Body (JSON)

Name	Type	Required	Description
email	email	Yes	Registered email address
code	string	Yes	6-digit verification code

Success Response

200

{"success": true, "message": "User verified successfully"}

Error Responses

400

Code expired / Incorrect verification code

404

User not found

POST

/users/auth/is_email_available/

Check if an email address is available for registration.

Request Body (JSON)

Name	Type	Required	Description
email	email	Yes	Email address to check

Success Response

200	{"success": true, "message": "Email is available"}
-----	--

Error Responses

400	Email is already used
-----	-----------------------

POST

/users/auth/verify_reset_email/

Verify email exists and send a reset code for the password reset flow.

Request Body (JSON)

Name	Type	Required	Description
email	email	Yes	Registered email address

Success Response

200	{"success": true, "message": "Email exists, verification code sent"}
-----	--

Error Responses

400	Email does not exist
-----	----------------------

POST

/users/auth/verify_phone/

Check if a phone number is available for registration.

Request Body (JSON)

Name	Type	Required	Description
phone	string	Yes	Phone number (+XXXXXXXXXXXX, 8-15 digits)

Success Response

200	{"success": true, "message": "Phone number is available"}
-----	---

Error Responses

400	Phone number is already used / Invalid phone format
-----	---

POST

/users/auth/forget_password/

Reset user password. Requires prior verification via verify_reset_code.

Request Body (JSON)

Name	Type	Required	Description
email	email	Yes	Registered email address
password	string	Yes	New password

Success Response

200	{"success": true, "message": "Password updated"}
-----	--

Error Responses

400	Email not verified for reset / Invalid email
-----	--

DELETE

/users/auth/{id}/delete_account/

Soft-delete a user account. Appends ".deleted" to email and phone. Requires authentication and ownership.

URL Parameters

Name	Type	Required	Description
id	integer	Yes	User ID (URL parameter)

Success Response

200

{"success": true, "message": "Account deleted successfully"}

Error Responses

400	Account already deleted
404	User does not exist
403	Permission denied (not owner or admin)
500	Internal server error

NOTE

Permission: IsAuthenticated + IsOwnerOrAdmin

02

FCM Devices

Firebase Cloud Messaging device registration for push notifications. Requires authentication.

BASE URL	/users/devices/
----------	-----------------

POST

/users/devices/register_device/

Register or update an FCM device token for the authenticated user.

Request Body (JSON)

Name	Type	Required	Description
fcm_token	string	Yes	Firebase Cloud Messaging device token

Success Response

200	{"success": true, "message": "Device registered successfully"}
-----	--

Error Responses

401	Authentication credentials not provided
-----	---

NOTE	Permission: IsAuthenticated. Uses update_or_create to avoid duplicates.
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03

Patients

Patient profile management including creation (linked to User), retrieval with computed fields (age, BMI, menstrual cycle), and profile updates.

BASE URL	/patients/PatientService/
----------	---------------------------

POST

/patients/PatientService/create_patient/

Create a patient profile linked to an existing User. For female patients, also creates a MenstrualCycle record.

Request Body (JSON)

Name	Type	Required	Description
email	email	Yes	Email of the linked User account
height	integer	Yes	Height in cm (min: 1)
weight	decimal	Yes	Weight in kg (min: 1)
blood_type	string	No	A+, A-, B+, B-, AB+, AB-, O+, O-
chronic_diseases	string	No	Chronic diseases description
allergies	string	No	Known allergies
current_medications	string	No	Current medications
family_doctor_name	string	No	Family doctor name (letters only)

Success Response

201	{"success": true, "message": "Patient created", "patient_id": 1}
-----	--

Error Responses

400	Invalid data or constraint violated / User not found
500	Database error

GET

/patients/PatientService/{id}/get_patient_by_id/

Get full patient profile with computed fields: age, BMI, menstrual cycle info (females only), and pregnancy week.

URL Parameters

Name	Type	Required	Description
id	integer	Yes	Patient ID (URL parameter)

Success Response

200

{"success": true, "patient": {...}, "age": 28, "bmi": 22.5, "menstrual_cycle": {...}}

Error Responses

- 404

Patient not found
- 400

Patient has incomplete information
- 500

Database error

NOTE

BMI, age, and pregnancy_week are computed server-side. menstrual_cycle is null for male patients.

PATCH

/patients/PatientService/{id}/update_patient/

Partially update patient profile fields. Only provided fields are updated.

URL Parameters

Name	Type	Required	Description
id	integer	Yes	Patient ID (URL parameter)

Request Body (JSON)

Name	Type	Required	Description
height	integer	No	Height in cm
weight	decimal	No	Weight in kg
blood_type	string	No	Blood type
chronic_diseases	string	No	Chronic diseases
allergies	string	No	Known allergies
first_name	string	No	Update user first name
last_name	string	No	Update user last name

Success Response

200	{"success": true, "message": "Patient updated"}
-----	---

Error Responses

404	Patient not found
400	No fields to update / Validation error
500	Database error

04

Doctors

Doctor profile management. Doctors are linked to User accounts and have specialities. Includes CRUD operations and listing all doctors.

BASE URL	/doctors/DoctorService/
----------	-------------------------

POST

/doctors/DoctorService/create_doctor/

Create a doctor profile linked to an existing User account.

Request Body (JSON)

Name	Type	Required	Description
user_id	integer	Yes	ID of the linked User account
speciality_id	integer	Yes	ID of the doctor speciality
ville	string	No	City (TUNIS, ARIANA, SOUSSE, etc.)
address	string	Yes	Full address
experience	integer	Yes	Years of experience (min: 0)
consultation_price	decimal	No	Consultation price
bio	string	No	Brief bio/description

Success Response

201	{"success": true, "message": "Doctor created successfully", "doctor_id": 1}
-----	---

Error Responses

400	Invalid data or constraint violated
500	Database error

GET

/doctors/DoctorService/{id}/get_doctor_by_id/

Get full doctor profile with computed age, speciality details, and user information.

URL Parameters

Name	Type	Required	Description
id	integer	Yes	Doctor ID (URL parameter)

Success Response

200

{"success": true, "doctor": {...}, "age": 35}

Error Responses

404

Doctor not found

500

Database error

NOTE

Age is computed from user birth_date. Returns null age if birth_date is missing.

PATCH

/doctors/DoctorService/{id}/update_doctor/

Partially update doctor profile. Can update both doctor fields and linked user fields.

URL Parameters

Name	Type	Required	Description
id	integer	Yes	Doctor ID (URL parameter)

Request Body (JSON)

Name	Type	Required	Description
ville	string	No	City
address	string	No	Full address
experience	integer	No	Years of experience
consultation_price	decimal	No	Consultation price
bio	string	No	Brief bio/description
first_name	string	No	Update user first name
last_name	string	No	Update user last name

Success Response

200

{"success": true, "message": "Doctor updated"}

Error Responses

404

Doctor not found

400

No fields to update / Validation error

500

Database error

GET

/doctors/DoctorService/get_all_doctors/

Retrieve a list of all registered doctors with their profiles and specialties.

Success Response

200

{"success": true, "doctors": [{...}, {...}]}

Error Responses

500

Database error

05

Measurements

Health measurement tracking and ML-powered risk assessment. Supports weight, blood pressure, glycemia, temperature, heart rate, and oxygen measurements. Uses atomic transactions.

BASE URL	/measurements/MeasurementService/
----------	-----------------------------------

POST

/measurements/MeasurementService/create_measurement/

Record a new health measurement. Automatically triggers ML risk assessment using all latest measurements. Sends email alert if risk is MEDIUM or HIGH. Wrapped in atomic transaction.

Request Body (JSON)

Name	Type	Required	Description
patient_id	integer	Yes	Patient ID
type	string	Yes	WEIGHT, BLOOD_PRESSURE, GLYCEMIA, TEMPERATURE, HEART_RATE, OXYGEN
value1	decimal	Yes	Primary value (e.g., systolic BP, weight)
value2	decimal	No	Secondary value (diastolic BP only)
unit	string	Yes	KG, MMHG, G_L, C, BPM
context	string	No	Additional context (default: "")

Success Response

200	{"success": true, "risk_level": "LOW", "risk_percentage": 15.2, "message": "Measurement created and risk assessment completed"}
-----	---

Error Responses

200	Measurement created but insufficient data for risk assessment (risk_level: null)
400	Invalid data or constraint violated
500	Database error

NOTE Uses transaction.atomic() - both Measurement and RiskAssessment are rolled back on failure.

GET

/measurements/MeasurementService/{id}/get_latest_measurements/

Get the latest measurement of each type for a patient, plus computed BMI and pregnancy week.

URL Parameters

Name	Type	Required	Description
id	integer	Yes	Patient ID (URL parameter)

Success Response

200	{"success": true, "height": 170, "weight": 65, "bmi": 22.5, "glycemia_informations": {...}, "blood_pressure": {.
-----	--

Error Responses

404	Patient not found
400	Patient has incomplete information (missing height/weight)
500	Database error

NOTE pregnancy_week is null for male patients or if no active pregnancy exists.

GET

/measurements/MeasurementService/{id}/get_patient_measurements/

Get all measurements for a patient, ordered by date (newest first).

URL Parameters

Name	Type	Required	Description
id	integer	Yes	Patient ID (URL parameter)

Success Response

200	{"success": true, "measurements": [{...}, {...}]}
-----	---

Error Responses

404	Patient not found
500	Database error

GET

/measurements/MeasurementService/{id}/get_risk_assessment/

Get the latest risk assessment for a patient (ML-predicted risk level and contributing factors).

URL Parameters

Name	Type	Required	Description
id	integer	Yes	Patient ID (URL parameter)

Success Response

200 {"success": true, "risk_assessment": {"global_risk_level": "LOW", "global_risk_percentage": 15.2, ...}}

Error Responses

404 Patient not found / No risk assessment found

500 Database error

06

Pregnancies

Pregnancy tracking for female patients. Manages pregnancy test results, due dates, and linked functionality with measurements (pregnancy week) and alerts.

BASE URL	/pregnancies/PregnancyService/
----------	--------------------------------

POST

/pregnancies/PregnancyService/create_pregnancy/

Record a new pregnancy entry for a patient.

Request Body (JSON)

Name	Type	Required	Description
patient	integer	Yes	Patient ID
test_date	date	Yes	Date of pregnancy test (YYYY-MM-DD)
test_result	boolean	Yes	Pregnancy test result (true/false)
start_date	date	No	Pregnancy start date
due_date	date	No	Expected due date
end_date	date	No	Pregnancy end date (null = ongoing)

Success Response

201

{"success": true, "message": "Pregnancy created successfully", "pregnancy_id": 1}

Error Responses

400

Invalid data or constraint violated

500

Database error

GET

/pregnancies/PregnancyService/{id}/get_current_pregnancy/

Get the current active pregnancy for a patient (where end_date is null).

URL Parameters

Name	Type	Required	Description
id	integer	Yes	Patient ID (URL parameter)

Success Response

200

{"success": true, "pregnancy": {"test_date": "...", "start_date": "...", "due_date": "...", ...}}

Error Responses

404

Patient not found / No active pregnancy found

500

Database error

PATCH

/pregnancies/PregnancyService/{id}/update_pregnancy/

Partially update pregnancy details (e.g., set end_date, update due_date).

URL Parameters

Name	Type	Required	Description
id	integer	Yes	Pregnancy ID (URL parameter)

Request Body (JSON)

Name	Type	Required	Description
start_date	date	No	Pregnancy start date
due_date	date	No	Expected due date
end_date	date	No	Set to close pregnancy

Success Response

200

{"success": true, "message": "Pregnancy updated successfully"}

Error Responses

404

Pregnancy not found

400

Validation error

500

Database error

DELETE

/pregnancies/PregnancyService/{id}/delete_pregnancy/

Permanently delete a pregnancy record.

URL Parameters

Name	Type	Required	Description
id	integer	Yes	Pregnancy ID (URL parameter)

Success Response

200

{"success": true, "message": "Pregnancy deleted successfully"}

Error Responses

404

Pregnancy not found

500

Database error

07

Alerts

Alert system for health notifications, medication reminders, appointment reminders, and system alerts. Most endpoints are triggered automatically by the APScheduler cron jobs. Frontend uses `get_alerts_by_user` and `mark_as_read`.

BASE URL	/alerts/AlertService/
----------	-----------------------

POST

/alerts/AlertService/send_risk_alert/

Send a health risk alert email and create an Alert record. Called automatically by the measurement service when risk is MEDIUM or HIGH.

Request Body (JSON)

Name	Type	Required	Description
email	email	Yes	Patient email address
alert_message	string	Yes	Alert message content
alert_level	string	Yes	WARNING or CRITICAL

Success Response

200

{"success": true, "message": "Alert sent successfully"}

Error Responses

- 400

Missing required fields (email, alert_message, alert_level)
- 404

User not found
- 500

Database error

NOTE

Automatic: Called by MeasurementService.createMeasurement() when risk detected.

POST **/alerts/AlertService/send_medication_reminders/**

Send medication reminder emails to all patients with active treatments due within the next hour. Creates Alert records.

Success Response

200 {"success": true, "message": "Medication reminders sent successfully"}

Error Responses

500 Database error

NOTE Automatic: Scheduled every hour at :00 via APScheduler.

POST **/alerts/AlertService/send_appointment_reminders/**

Send appointment reminder emails to patients with confirmed appointments tomorrow.

Success Response

200 {"success": true, "message": "Appointment reminders sent successfully"}

Error Responses

500 Database error

NOTE Automatic: Scheduled daily at 18:00 via APScheduler.

POST **/alerts/AlertService/send_missing_measurements_alerts/**

Alert patients who haven't recorded any measurements in the last 3 days.

Success Response

200 {"success": true, "message": "Missing measurements alerts sent successfully"}

Error Responses

500 Database error

NOTE Automatic: Scheduled daily at 09:00 via APScheduler.

POST

/alerts/AlertService/send_unconfirmed_appointment_alerts/

Alert patients and doctors about appointments in the next 2 days that are still PENDING.

Success Response

200

{"success": true, "message": "Unconfirmed appointment alerts sent"}

Error Responses

500

Database error

NOTE

Automatic: Scheduled daily at 09:15 via APScheduler. Sends to BOTH patient and doctor.

POST

/alerts/AlertService/send_pregnancy_no_appointment_alerts/

Alert pregnant patients who have no upcoming appointments scheduled.

Success Response

200

{"success": true, "message": "Pregnancy no appointment alerts sent"}

Error Responses

500

Database error

NOTE

Automatic: Scheduled daily at 09:30 via APScheduler.

GET

/alerts/AlertService/{id}/get_alerts_by_user/

Get all alerts for a specific user, ordered by creation date (newest first). For frontend use.

URL Parameters

Name	Type	Required	Description
id	integer	Yes	User ID (URL parameter)

Success Response

200

{"success": true, "alerts": [{"type": "HEALTH", "message": "...", "level": "...", "status": "NEW", ...}]}

Error Responses

404

User not found

500

Database error

NOTE

Frontend endpoint: Use this to display alerts in the app.

PATCH

/alerts/AlertService/{id}/mark_as_read/

Mark a specific alert as read by changing its status from NEW to READ.

URL Parameters

Name	Type	Required	Description
id	integer	Yes	Alert ID (URL parameter)

Success Response

200

{"success": true, "message": "Alert marked as read"}

Error Responses

404

Alert not found

500

Database error

NOTE

Frontend endpoint: Call when user opens/views an alert.

08

Medications

Medication catalog import, DCI linking, treatment scheduling, and search endpoints.

BASE URL	/medications/medicationsService/
----------	----------------------------------

POST

/medications/medicationsService/create_medications/

Load medications from the sources/MED_CNAM.csv file into the database.

Success Response

201	{"message": "Medications created successfully"}
-----	---

Error Responses

404	CSV file not found
400	Invalid data / Database integrity error
500	Database error / Server error

NOTE No request body. Intended for initial data load.

POST

/medications/medicationsService/create_medications_dci/

Create Medication-DCI relationships using sources/special_DCI.txt and existing medications.

Success Response

201	{"message": "Medication-DCI relationships created successfully"}
-----	--

Error Responses

207	Completed with some unmatched DCIs
404	special_DCI.txt file not found
400	Database integrity error
500	Database error / Server error

NOTE No request body. Returns 207 if some DCIs are unmatched.

POST

/medications/medicationsService/create_treatment_with_schedules/

Create a treatment and its schedules in a single request.

Request Body (JSON)

Name	Type	Required	Description
treatment	object	Yes	Treatment details
treatment.patient_id	integer	Yes	Patient ID
treatment.medication_id	integer	Yes	Medication ID
treatment.start_date	date	Yes	Start date (YYYY-MM-DD)
treatment.end_date	date	No	End date (optional)
treatment.dose	string	Yes	Dose description
treatment.frequency	string	Yes	Frequency description
schedules	array	Yes	List of schedules
schedules[].dose_time	time	Yes	Dose time (HH:MM[:SS])

Success Response

201

{"success": true, "message": "Treatment created successfully", "treatment": {...}, "schedules": [...]}

Error Responses

400

Invalid data or constraint violated

500

Database error / Server error

GET

/medications/medicationsService/{id}/get_treatments_by_patient_id/

Retrieve all treatments for a patient, including schedules.

URL Parameters

Name	Type	Required	Description
id	integer	Yes	Patient ID (URL parameter)

Success Response

200

{"success": true, "treatments": [{...}, {...}]}

Error Responses

404

No treatments found for this patient

PATCH

/medications/medicationsService/{id}/update_schedule_by_id/

Update the dose time for a specific treatment schedule.

URL Parameters

Name	Type	Required	Description
id	integer	Yes	Schedule ID (URL parameter)

Request Body (JSON)

Name	Type	Required	Description
new_time	time	Yes	New dose time (HH:MM[:SS])

Success Response

200	{"success": true, "message": "Schedule updated successfully", "schedule": {...}}
-----	--

Error Responses

400	new_time is required
404	Schedule not found
500	Server error

DELETE

/medications/medicationsService/{id}/delete_treatment_by_id/

Delete a treatment by its ID.

URL Parameters

Name	Type	Required	Description
id	integer	Yes	Treatment ID (URL parameter)

Success Response

200	{"success": true, "message": "Treatment deleted successfully"}
-----	--

Error Responses

404	Treatment not found
500	Server error

DELETE

/medications/medicationsService/{id}/delete_schedule_by_id/

Delete a treatment schedule by its ID.

URL Parameters

Name	Type	Required	Description
id	integer	Yes	Schedule ID (URL parameter)

Success Response

200

{"success": true, "message": "Schedule deleted successfully"}

Error Responses

404

Schedule not found

500

Server error

GET

/medications/medicationsService/{id}/get_medication_by_id/

Get medication details by ID. For female users, includes pregnancy risk data if an active pregnancy exists.

URL Parameters

Name	Type	Required	Description
id	integer	Yes	Medication ID (URL parameter)

Success Response

200

{"success": true, "medication": {...}, "pregnancy_data": {...}}

Error Responses

401

Authentication required

404

User not found / Medication not found / Patient not found / No active pregnancy found

400

Unable to determine current trimester

NOTE

Requires authentication. pregnancy_data is null when no DCI links exist or user is not eligible.

GET

/medications/medicationsService/search/

Search medications by name, DCI, dosage, or form. Supports fuzzy search and code lookup.

URL Parameters

Name	Type	Required	Description
q	string	Yes	Search query (min 2 characters)

Success Response

200

{"count": 3, "results": [{...}, {...}]}

Error Responses

400

Query must be at least 2 characters

404

No medications found

09

DCI

DCI (drug classification) import endpoint based on sources/classifications.csv.

BASE URL	/dci/DCI/
----------	-----------

POST

/dci/DCI/create_dci/

Load DCI classifications from CSV and store them in the database.

Success Response

201	{"message": "DCIs created successfully"}
-----	--

Error Responses

404	CSV file not found
400	Invalid data / Database integrity error
500	Database error / Server error

NOTE No request body. Intended for initial data load.

Additional Endpoint: Token Refresh

POST /users/refresh/

Body: {"refresh": "<refresh_token>"} -> Returns: {"access": "<new_access_token>"} | SimpleJWT built-in view